Kubernetes as a Virtual Machine Management System

A Comparative Study of KubeVirt and Traditional Approaches

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Abstract

Sammanfattning

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Chapter 1

Introduction

1.1 Background

Virtualization is a technique that either splits or combines computing resources into one or more operating environments by creating a software layer that surrounds or underlies existing hardware or software. This enables the creation of virtualized resources that are decoupled from the underlying hardware. Virtualization can occur in various forms including the virtualization of storage, networks and servers. Virtualizing networks and storage is achieved by creating an abstraction layer on top of the physical network and storage resources, respectively, to provide a more flexible and scalable environment for other resources to interact with. Server virtualization, on the other hand, is used to create virtual computing environments that are decoupled from the underlying computing resources. By combining virtualized resources into a single operating environment, virtualization allows heterogeneous hardware resources to be used homogeneously and thus it improves resource utilization and flexibility [1, 2, 3].

To enable server virtualization and manage virtual computing environments, a hypervisor is used. A hypervisor is a software layer that operates between the hardware and the operating system (OS) and is responsible for creating virtual computing environments called virtual machines (VMs) [3, 4, 5]. VMs are emulations of physical computers that are perceived as real computers by applications running inside them. They provide a complete and independent environment by virtualizing an OS, rendering them suitable for a wide range of applications, including legacy applications and applications that require a high level of isolation. For this reason, VMs have long been the core part of IT infrastructure, serving as foundational components in system

architecture [1, 2].

In parallel with the evolution of VMs, containers have emerged as an alternative to VMs. Container virtualization is an OS-level virtualization technique that allows for running multiple isolated applications on a single OS instance. This means container virtualization does not require a hypervisor, and instead uses the features and capabilities of the OS kernel to create and manage containers. As such, containers are more lightweight, require less overhead, and have faster startup times than VMs [6]. This has made containers a popular choice for scenarios in which scaling and portability are key, such as in microservices, which are core components in modern cloud-native applications [7]. Figure 1.1 visualizes the differences between how VMs and containers are virtualized, and how the virtualization layer is placed on a lower level for VMs than for containers.

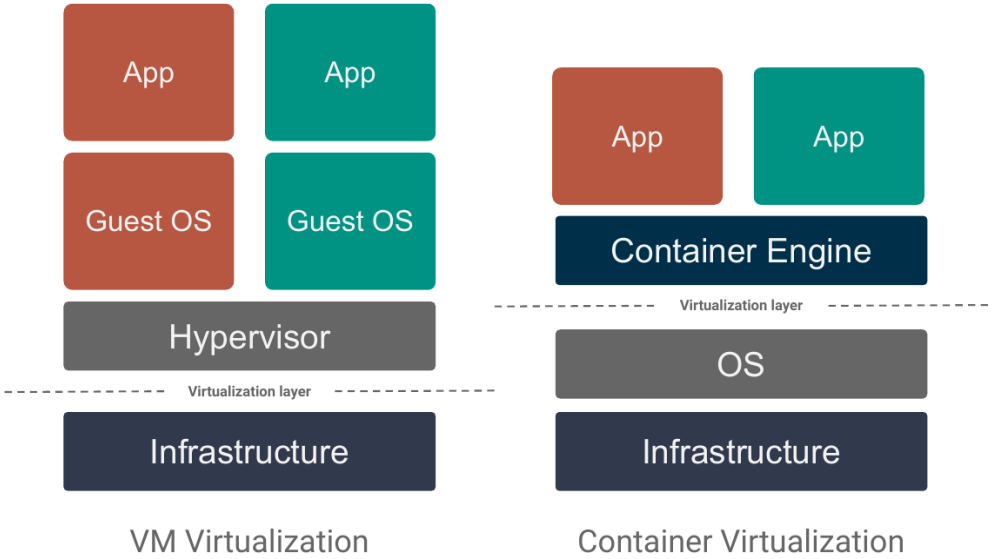

Figure 1.1: Overview of VMs and containers

However, while containers provide a lightweight and efficient alternative to VMs, they lack the deep isolation properties inherent to VMs. Sharing the OS kernel with the host and other containers opens up a larger attack surface, which implies that containers are more prone to security vulnerabilities. While there exist technologies that aim to unify the isolation benefits of VMs and the lightweight benefits of containers, such as Kata Containers, gVisor and Firecracker, it remains a fact that both containers and VMs have their use cases [8, 9].

Due to their different use cases, VMs and containers have traditionally been

managed separately. VMs are often managed by a VM management system that provides a set of tools and processes to aid in the life-cycle management of VMs. This makes a VM management system act as a higher-level abstraction layer that unifies hypervisors, typically through an application programming interface (API) such as a command line interface (CLI) or a web-based user interface. Figure 1.2 depicts a broad overview of the role of a VM management system where clients interact with an interface to manage VMs that are distributed across several hosts. Moreover, a VM management system is often part of a larger infrastructure, such as Infrastructure as a Service (IaaS), that provides a wider range of virtualized resources including virtual networks and storage. Examples of such systems include OpenStack [10], CloudStack [11], and OpenNebula [12]. Containers, on the other hand, are often managed by a container orchestration platform. These platforms provide a similar set of tools and processes as a VM management system but for containers. Apart from creating and deploying containers, a container orchestration platform provides tools specifically designed to manage the life cycle of containers, such as scheduling and scaling. Examples of such platforms include Kubernetes [13], Nomad [14], and Docker Swarm [15], with the former being the most widely adopted [16]. Furthermore, with the evolution and adaption of container orchestration platforms like Kubernetes, there has been an ongoing convergence in how VMs and containers are managed [17].

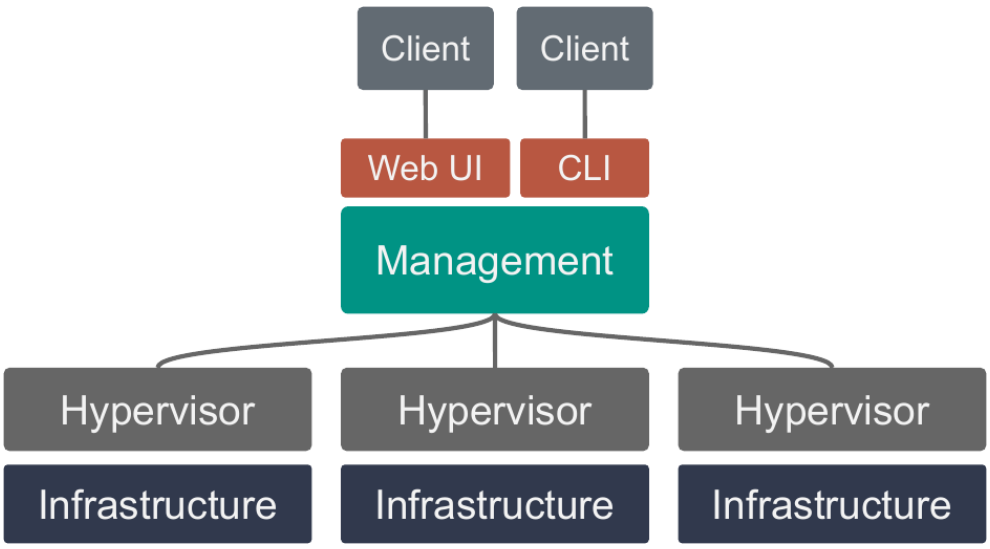

Figure 1.2: Overview of a VM management system

Kubernetes, originally designed to manage containerized applications at scale, has recently been extended to accommodate VMs through the KubeVirt

project. KubeVirt is an API extension for Kubernetes, that enables the running of VMs inside a Kubernetes environment alongside containers, creating a common environment for both virtualization technologies [18]. Figure 1.3 depicts a high-level overview of how VMs and containers coincide in a Kubernetes environment using KubeVirt, where traffic flows natively between the two types of resources. Further technical details of how KubeVirt achieves this will be discussed in 2.6.1.

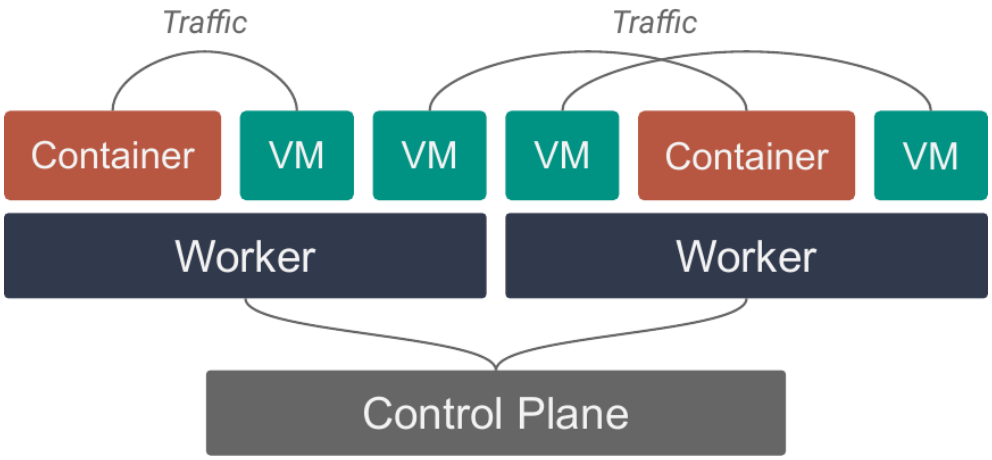

Figure 1.3: Overview of VMs and containers managed by using KubeVirt

The approach of incorporating VMs in Kubernetes not only benefits from the flexible, scalable and mature infrastructure that Kubernetes provides but also from tools and processes that are designed to integrate with or manage the life cycle of Kubernetes resources. This means by using KubeVirt, VMs can be managed similarly to containers, creating a more cloud-native approach to VM management [7]. Furthermore, consolidating the management of VMs and containers into a single platform could offer benefits in terms of reduced architectural complexity and simpler hardware management, but also practical aspects such as reduced required skill set in an organization [19, 20].

The integration of VM management into Kubernetes through KubeVirt represents a significant milestone toward a more agile, efficient, and unified IT infrastructure [20]. However, KubeVirt's practicality in a real-world scenario is yet to be fully assessed. As such, this project aims to evaluate Kubernetes with KubeVirt as a VM management system and compare it with traditional approaches to understand its practicality and limitations.

1.2 Research Questions

What performance, scalability, and operational efficiencies can be gained by integrating VMs in Kubernetes?

This could stem from using a single resource scheduler. Several metrics will be used to evaluate this, as described in the background and related work.

What are the current challenges and limitations in managing VMs and containerized applications in separate environments?

This is concerned with scenarios in which a traditional VM management system is operated alongside Kubernetes.

How can system administration be simplified by running VMs in Kubernetes using KubeVirt?

This is focused on the practical implications of using KubeVirt and Kubernetes instead of traditional VM management systems. For instance, the number of steps required to deploy a VM.

1.3 Problem

Due to the different use cases for VMs and containers, they have traditionally been managed separately. However, both VMs and containers are still part of the necessary infrastructure many organizations depend on. This has led to fragmented infrastructure with separate management systems and tools to manage VMs and containers with a direct implication of increased architectural complexity and a larger required skill set for operators. Furthermore, many cloud-native technologies are designed for Kubernetes, and as such, by running VMs outside of Kubernetes, these benefits might not be utilized.

1.4 Purpose

The purpose of this project is to evaluate if Kubernetes, a platform originally designed for container orchestration, is a suitable platform for managing VMs using KubeVirt. This evaluation is important in the field of cloud computing, as it could provide insights into how the management of VMs and containers

could be both simplified and unified, but also how the benefits of Kubernetes can be utilized for VMs.

1.5 Goals

This project aims to explore the integration of VMs into Kubernetes environments using KubeVirt. By comparing how VMs are managed using KubeVirt against traditional VM management systems, this project aims to address possible inefficiencies and complexities that arise when managing VMs outside of Kubernetes. The project will investigate the implications of moving into a KubeVirt-based system, such as the benefits and limitations of using KubeVirt, and how it compares to traditional VM management systems.

1.6 Research Method

This project will use a mixed-methods approach to investigate the research questions. Initially, a literature review will be carried out to understand the current state of the art in VM management systems and the current challenges faced and limitations in managing VMs and containers in separate environments. Following this, both a qualitative and quantitative comparison will be conducted to understand the practical implications of moving into a KubeVirt-based system. The qualitative comparison will be conducted to understand the practical implications of moving into a KubeVirt-based system. It will include collecting feedback by interviewing system administrators and operators with experience in both traditional VM management systems and KubeVirt. The quantitative comparison will be conducted to evaluate measurable performance, scalability, and operational efficiencies that could arise when using Kubernetes with KubeVirt, and will include a set of benchmarks.

1.7 Limitations

1. This project will not investigate any solutions for shortcomings in KubeVirt itself, such as finding alternatives for features that exist in existing virtual machine management systems but not in KubeVirt. As such, this project acknowledges that KubeVirt might not be a complete replacement for traditional virtual machine management tools since it might not offer all the features of an IaaS system. Therefore, only features that

exist in KubeVirt and the compared system will be considered when testing.

2. This project will primarily focus on the practical implications of moving into a KubeVirt-based system. For this, there are mainly two scenarios that will be considered:

   - **Replacement**: A system uses only a traditional virtual management system. By using KubeVirt, the system will be replaced by a Kubernetes-based system.
   - **Unifying**: A system uses both traditional management and Kubernetes. By using KubeVirt, the system will be unified into a single platform for both VMs and containers.

1.8 Structure of the thesis

Chapter 2

Background

This chapter will discuss the relevant background information and concepts that are necessary to understand why KubeVirt has emerged and how it fits into the current landscape of VMs and containers. It is structured linearly, starting with a detailed discussion of virtualization, hypervisors and VMs, followed by a comparison of VM management systems. It then introduces containers and Kubernetes, and how they have emerged as an alternative to VMs, and finally discusses the integration of VMs into Kubernetes through KubeVirt.

2.1 Virtualization Fundamentals

Virtualization is a technology that enables partitioning or grouping computing resources into operating environments using methods such as partitioning, emulation, and time-sharing [2]. Virtualization can occur in various forms including the virtualization of storage, networks and servers. Storage virtualization is achieved by creating an abstraction layer on top of physical storage resources to provide a single interface to storage resources that appear as a single storage device. This allows for a more flexible environment where storage resources can be added or removed without affecting the application layer. Similarly, network virtualization enables the creation of networks with the properties of a physical network, but with its appliances such as routers and switches being implemented in software. This allows for flexibility when managing resources in a network since common practices such as networking partitioning and isolation can be achieved without acquiring additional hardware. Server virtualization, on the other hand, is concerned with creating virtual computing environments that are decoupled from the underlying computing resources. It is often referred to as hypervisor-based virtualization, as

it requires a software layer called a hypervisor that is responsible for creating and managing the virtual computing environments [1].

A virtual computing environment, more often referred to as a virtual machine (VM), is an emulation of a physical computer and is perceived as a *real* computer by the application running inside it even though the underlying resources are often shared with other VMs. To enable this, a hypervisor divides the hardware into virtualized resources, such as virtual CPUs and virtual memory, and provides an interface to manage these resources. For instance, an application running inside a VM is unaware that the resources it is using are virtualized, and instead interacts with the OS and hardware as if it were running on a physical machine. This is particularly useful for scenarios in which hardware is under-utilized, as it allows for the consolidation of workloads onto a smaller number of physical machines, which in turn reduces the need for acquiring additional hardware. [2, 3].

Depending on the required level of abstraction, virtualization can be achieved in different layers, where the three major being hardware-level, OS-level and application-level. For instance, storage and network virtualization both fall under the category of hardware virtualization, as they create an abstraction layer on top of the physical resources. However, in other scenarios, virtualization is not required to be as low-level, such as in application virtualization. Application virtualization is instead concerned with encapsulating an application in a lightweight virtualized environment, such that it is perceived as being run on a specific platform. A common example is the Java Virtual Machine (JVM) that enables programs written in the programming language Java to be run on multiple operating systems [21]. Figure 2.1 presents an overview of the differences between the two more common layers of virtualization; hardware and OS virtualization. While the figure does not include application-level virtualization, it is important to note that the virtualization layers are not mutually exclusive and that they can be used in combination. For instance, combining hardware-level and OS-level virtualization creates a flexible environment where a guest process can run inside a guest OS [2].

Furthermore, virtualization can be divided into two categories, full virtualization and paravirtualization. Full virtualization is the traditional approach where the entire set of hardware is emulated, and the VM is presented with an indistinguishable image of the underlying hardware. This allows for complete decoupling of the VM from the underlying resources and does not require any modification to the guest OS [2, 22, 23]. Paravirtualization, on the other hand, is a subset of full virtualization that aims to improve VM performance by reducing the number of resources that are emulated. The VM is presented with

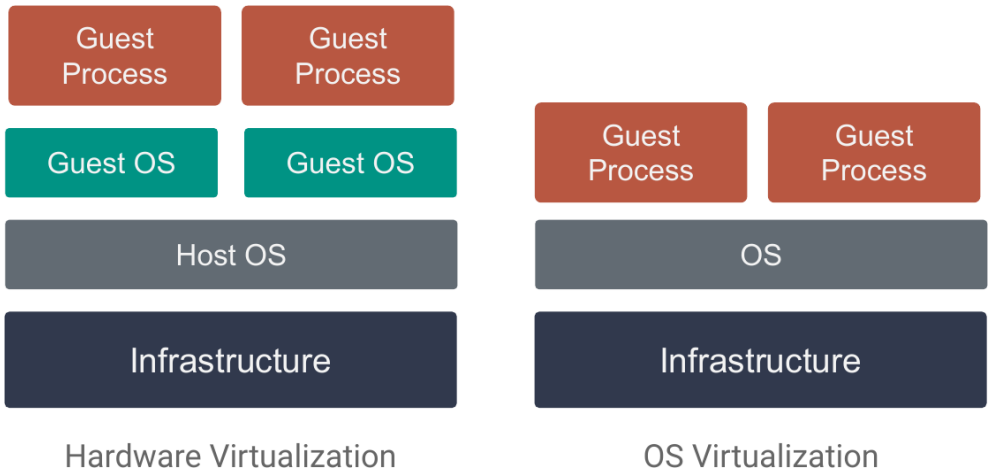

Figure 2.1: Overview of the different layers of virtualization

an image that is similar, but not identical, to the underlying hardware. As such, the guest OS is aware it is running inside a VM [22, 23].

Benefits and Challenges

Many of the benefits of virtualization stem from the fact that there exists an abstraction layer above the physical hardware. The abstraction layer decouples software from hardware, allowing for the use of vendors with different I/O in the same environment. This creates a more flexible and scalable computing environment as a whole where hardware is turned into a simple pool of computing resources [3]. As such, there are numerous benefits of virtualization, a few of which are listed below:

- Resource utilization: Virtualization allows for better utilization of hardware resources since multiple VMs can run on the same physical hardware. This is achieved by sharing the underlying resources between the VMs, such as CPU and memory [2].

- Isolation: VMs can isolate their entire state from one another, including the operating system (OS), filesystem and application stack. This is particularly useful for certain applications with stricter security requirements, such as applications managing sensitive or confidential data. Isolation is also beneficial in scenarios where reproducibility is key, such as in testing and development environments [24].

- Flexibility: VMs are not tied to specific hardware, and can be moved

between different physical resources. Flexibility is an especially useful factor when in scenarios where hardware is not fully utilized since VMs can be migrated to other physical machines to balance the load [2].

- Management: VMs can be created and destroyed on demand, and can be managed using software tools [2].
- Virtual hardware: A VM can be configured with hardware that does not exist in the physical environment, such as a virtual network card or a virtual hard drive. This allows for better support for legacy applications or certain applications that are dependent on a specific hardware configuration [2].

However, the portability made possible by virtualization comes at a cost of performance. Since a VM is a complete but emulated system, every instruction issued must be interpreted in software. This means that despite the level of abstraction layer used, there is an inescapable added overhead when using virtualization. A common hypervisor such as KVM (see 2.2) introduces an overhead of 5 to 10% [25], and the overhead can be even larger when using a full virtualization approach.

2.2 Understanding Hypervisors

A hypervisor is a software layer that enables managing multiple VMs on a single physical machine by allowing multiple operating systems (called *guests*) to share the same hardware [1, 5]. Hypervisors, often referred to as virtual machine monitors (VMMs) are responsible for creating and managing VMs and for providing the necessary abstractions to the VMs.

Hypervisors can be categorized into two types, *type 1* and *type 2*, based on their operational context and how they interact with the underlying hardware. While the goals and purpose of type 1 and type 2 hypervisors are the same, there are some core differences. Type 1 hypervisors, commonly referred to as *bare-metal* hypervisors, run directly on top of the physical hardware and do not require an OS underneath. A type 1 hypervisor is therefore scalable to terabytes of RAM and hundreds of CPU cores [26]. However, without any other additional software, it is also responsible to schedule and allocate required system resources to the VMs. Type 2 hypervisors, also known as *hosted hypervisors*, rely on an underlying OS to manage the calls to the CPU and memory resources. The presence of an underlying OS inevitably introduces latency, which is why type 2 hypervisors are often used for development and testing

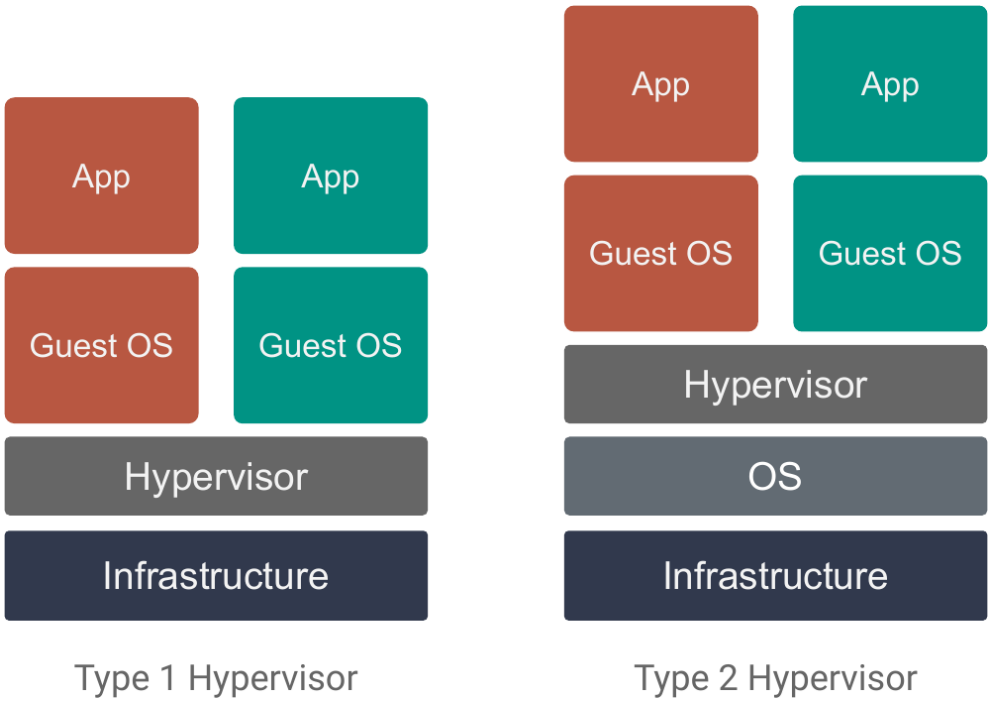

Figure 2.2: Overview of type 1 and type 2 hypervisors

purposes [5]. Figure 2.2 presents a high-level overview of the difference between a type 1 and type 2 hypervisor [5, 26].

Furthermore, hypervisors can be further categorized into *monolithic* and *microkernelized* hypervisors [27]. Monolithic hypervisors are built as a single large software stack, managing all the VMs and their resources, and hosting driving for all hardware access. Having all drivers reside within the hypervisor means that the hypervisor will only run on selected hardware. However, it also means no operating system is required at all, as the hypervisor acts as an operating system by supporting all the operations of its VMs. Microkernelized hypervisors, on the other hand, do not require device drivers to be present in the hypervisor. Instead, the drivers are installed in a form of managing VM, often referred to as parent VM. Each VM created by such a hypervisor is thus referred to as a child VM, as it uses the parent VM's drivers to access the hardware. This approach is more flexible, as it can allow the hypervisor to run on any hardware that supports the parent VM's operating system.

In the realm of hypervisors, the concept of *security rings* is often used to describe the level of privilege different part of a hypervisor has. A security ring represents a hierarchical protection domain, where the innermost ring (ring 0), usually reserved for the OS kernel, has the highest privilege, and the

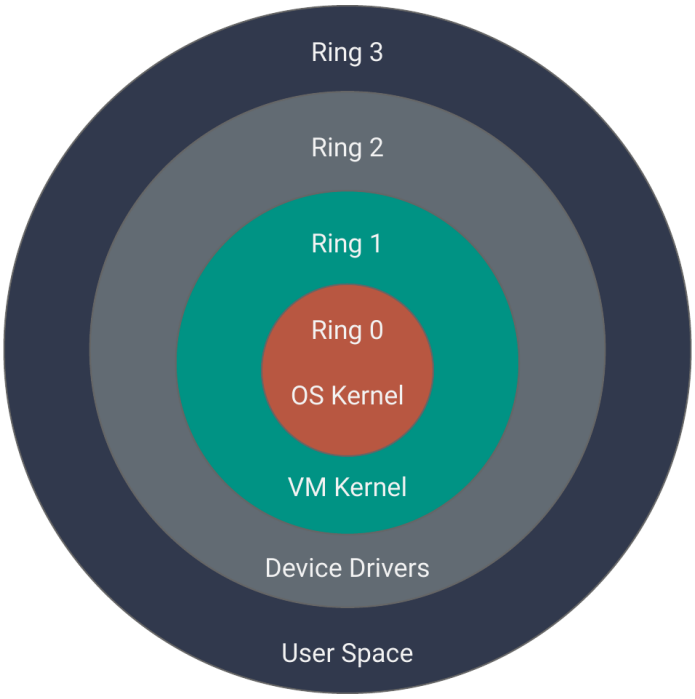

Figure 2.3: Overview of security rings in a hypervisor

outermost ring (ring 3), usually reserved for user applications, has the lowest privilege. The hypervisor itself is often placed in ring 0, its VM kernel space in ring 1, and the VM user space in ring 3. Security rings are thus used to define execution level to provide a system to limit the execution capabilities of VMs to regulate access to shared memory. This enabled a strong isolation layer between the VMs, and between the VMs and the hypervisor, thus a key part of a hypervisor's security model [27, 28]. Hypervisor security rings are visualized in figure 2.3.

Key Hypervisors

There are several hypervisors available, each with its own set of features and capabilities, some of which are listed below:

- VMware ESXi: A lightweight type 1 monolithic hypervisor often used in enterprise environments. ESXi is known for its ultra-lightweight footprint stretching merely a few hundred megabytes, and its ability to run directly on the hardware without an underlying OS [27].
- Microsoft Hyper-V: A type 1 microkernelized hypervisor that is part of

the Windows Server operating system. As a microkernelized hypervisor, Hyper-V differentiates between parent and child VMs and uses the parent VM to manage hardware access for the child VMs [27, 29].

- Citrix Hypervisor: Formerly known as XenServer, is a type 1 microkernelized hypervisor based on the Xen Project. It runs using a modified Linux kernel in a VM, called Dom0, that manages hardware I/O and interacts with the other virtual machines, called DomU guests [23, 27].
- KVM: A type 1 hypervisor that is part of the Linux kernel. As such, KVM uses most of the functionalities from the Linux kernel including the scheduler and memory management. KVM has become a popular alternative to proprietary hypervisors, such as VMware ESXi and Microsoft Hyper-V since it is open-source and free to use [27, 30].

2.3 VM Management Essentials

A management system for VMs involves a set of tools and processes that are used to create, deploy, and manage VMs. The management system is responsible for managing the entire life cycle of a VM, including provisioning, deployment, monitoring, and scaling. Not be confused with a hypervisor, a VM management system operates on a higher level and aims to provide an abstraction layer in the form of an interface to allow interaction with the VMs, often enabling the support of a set of hypervisors [31]. As such, a VM management system unifies a set of hypervisors by allowing them to be managed as if they were homogeneous, typically through a higher-level API, such as a CLI or a web-based user interface. However, certain hypervisor features, such as live migration, could be implemented differently across hypervisors, and it is up to the VM management system to provide a unified interface to manage such features [32].

Concepts and Features

With a higher abstraction layer, a VM management system brings a set of concepts to simplify the management of VMs. Many such features and concepts exist and can be broadly grouped as clustering, availability, and scalability [31].

Clustering involves grouping a set of resources, such as VMs or storage nodes, to provide an image that is perceived as a single resource. By allowing multiple nodes to cooperate and share resources, clustering can allow for

a more scalable and performant system. High-performance computing (HPC) is a common example of clustering where computing resources are aggregated to gain performance greater than that of a single computer. This enables HPC clusters to process large datasets and complex calculations that would take an infeasible amount of time to process on a single computer. Moreover, HPC clusters are designed such that they allow for the distribution of computational tasks where larger tasks are divided into smaller subtasks. However, the splitting of tasks is often not transparent to the end user, and the user perceives the cluster as a single resource [33, 34].

Availability refers to the ability to keep a resource active or running, which is, more often than not, clustered resources. A common practice for cloud computing is managing availability zones (AZs). An AZ groups resources such that each AZ is independent of the other AZs while providing the same service. This allows for outages to be isolated to a single AZ while the other AZs continue to provide the service. Thus, through redundancy and isolation, AZs enable high availability and fault tolerance [31, 35]. Figure 2.4 presents an overview of how 3 AZs provide redundant access to the same service.

Scalability involves technologies to ensure resources can manage an increase or decrease in computational loads. A VM management system often facilitates both scaling up and scaling out. Scaling up, also referred to as vertical scaling, involves increasing or decreasing the performance specification of a resource. For instance, a VM can be scaled up by increasing the number of CPUs or the amount of memory. Scaling out, also referred to as horizontal scaling, involves adding or removing entire resources, such as VMs. This means, that instead of upgrading a single VM, additional VMs are added to the system to handle the increased load. This approach is often regarded as more scalable than scaling up, as it allows for a more granular approach to resource management, and can be down without any downtime. Additionally, by avoiding downtime, the system appears as more available to the end user [36, 37]

Cloud Service Models

A VM management system is often part of a larger cloud computing infrastructure, such as Infrastructure as a Service (IaaS). IaaS is a lower-level cloud service model that aims to eliminate the need to install and manage physical hardware by providing virtualized computing resources, such as VMs, virtual networks and storage. As such, an IaaS platform provides a single interface to manage a wide range of virtualized resources that acts as a foundation for

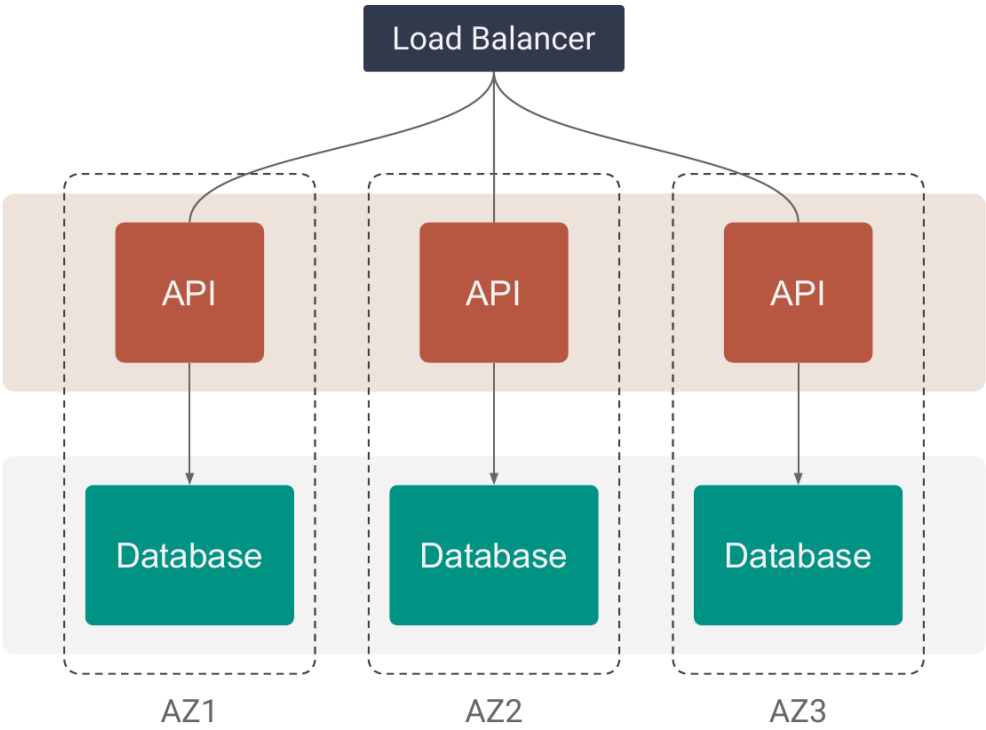

Figure 2.4: Overview of availability zones

building and managing cloud environments [38].

Other cloud service models include Platform as a Service (PaaS) and Software as a Service (SaaS). They provide a higher level of abstraction than IaaS forming a hierarchical relationship with each other. This means a SaaS service can be built on top of a PaaS service, which in turn can be built on top of an IaaS service. While figure 2.5 presents an overview of these cloud service models, both PaaS and SaaS are out of the scope of this project [39].

2.3.1 Comparison of VM Management Systems

Managing VMs can be done in many ways, and there are numerous VM management systems available, each with its own set of features and capabilities. Many such systems are a part of a larger platform, such as an IaaS or even as large as a cloud platform. Some of the most prevalent open-source platforms that have been used in the industry are OpenStack, CloudStack, and OpenNebula, and some of the most popular public cloud platforms are Amazon EC2, Azure Virtual Machines, and Google Compute Engine.

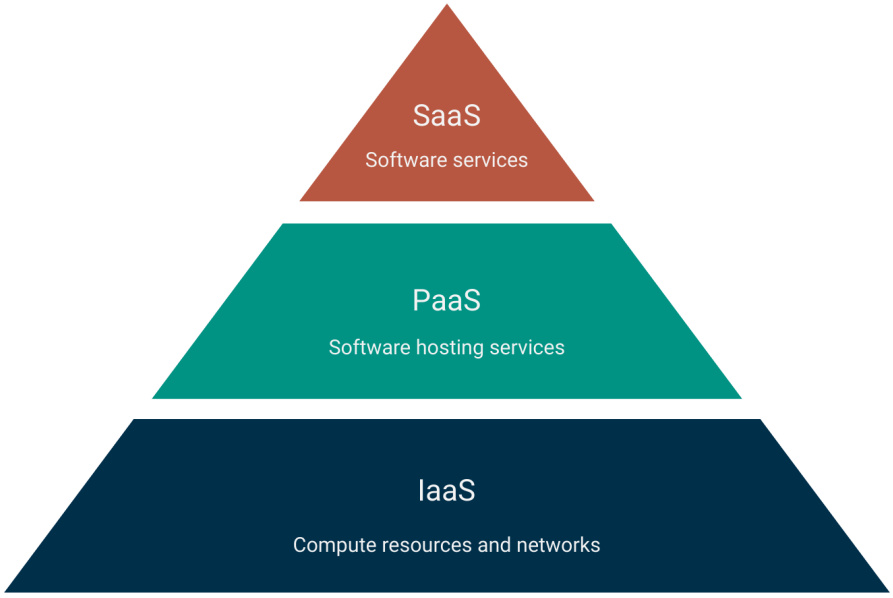

Figure 2.5: Overview of cloud service models

OpenStack

An open-source IaaS platform that offers a large set of features to manage VMs, networks, and storage [32]. OpenStack's architecture is inherently modular, and it is designed to be deployed in a distributed manner [40]. The architecture includes three major parts:

- *Nova*: The compute service that manages VMs and provides an interface to manage VMs. Nova supports all operations regarding the life cycle of a VM, such as creating, deleting, and migrating VMs. Nova is itself divided into multiple components, each employing a *shared-nothing* and *messaging-based* policy, which means every component is independent and communicates with other components using a message queue.

- *Swift*: The object storage service that provides a scalable and redundant storage system for VMs. It is a long-term storage for static data, such as images and backups, and is comparable to Amazon S3.

- *Glance*: The image service that provides a catalog of VM images that can be used to create new VMs.

OpenStack supports features such as live migration, snapshot management, and scalability, and is often used in large-scale cloud environments [32]. How-

ever, these features are often hypervisor-specific and require both support from the hypervisor and specific configuration to work [41].

CloudStack

An open-source IaaS platform and cloud computing software that manages computing, networking and storage resources, and is used to build and manage public and private cloud environments [42]. On a high level, CloudStack is built in a monolithic fashion, encapsulating all the management features into two software packages called the management server and agent. The management server is responsible for managing the life cycle of VMs, including provisioning, deployment, and scaling, and is comparable to a control node. The agent, however, is installed on each node designated to running VMs and is responsible for low-level management tasks of the VMs on that node, such as interacting with the underlying hypervisor. While it is a monolithic system, CloudStack is designed to be horizontally scalable and can be deployed in a distributed manner with multiple management servers [42].

CloudStack supports a large variety of hypervisors, such as VMware, KVM, and XenServer, and storage services, such as Amazon S3, allowing it to function as an abridgment layer for computing, network, and storage to unify a wide range of existing infrastructure [42]. Features such as live migration, snapshot management, and scalability are supported but are often hypervisor-specific and require specific configuration to work [43].

OpenNebula

An open-source IaaS platform and toolkit used to provide open, scalable, and comprehensive management of virtualized data centers [32]. OpenStack employs a modular architecture divided into three layers [44, 45]:

- *Tools layer*: Provides a set of interfaces such as a CLI and web-based user interface to interact with the system, but also high-level management tools such as scheduling.
- *Core layer*: Provides the core functionalities of the system, such as host management, VM management, and virtual network management.
- *Driver layer*: Provides an abstraction on top of the underlying infrastructure to enable the system to support a wide range of hardware, including networking, storage, and hypervisors. As such, the driver layer

is responsible for allocation, deallocation and monitoring of resources, such as VMs and storage.

By dividing the system into these layers, OpenNebula provides modularity that enables the system to be extended and customized to fit specific needs, such as integrating with other systems or using OpenNebula to build a private cloud.

Amazon EC2

Amazon Elastic Compute Cloud (EC2) is a web service that provides on-demand compute resources over the cloud [46, 47]. EC2 is part of Amazon Web Services (AWS), is designed to be scalable and flexible, and is often used to host general-purpose applications, including web applications, databases and enterprise applications. While the EC2 platform uses Xen-based hypervisors (see 2.2) under the hood [48], it acts as a complete abstraction layer over the underlying hardware and hypervisors and instead provides a rich set of tools to manage the VMs.

EC2 provides a large set of pre-configured templates called Amazon Machine Images (AMIs) that can be used to create new VMs, as well as a configuration in terms of specifications. This enables EC2 to provide a wide range of VMs, such as general-purpose, compute-optimized, memory-optimized, and storage-optimized VMs [46]. Features such as live migration, snapshot management, and auto-scaling are supported,

Azure Virtual Machines

Azure Virtual Machines is a service provided by Azure to manage VMs in the cloud [49], and is designed to be a cost-effective alternative to on-premises hardware [50]. Azure Virtual Machine provides a set of preconfigured VMs that can be used to create VMs for specific purposes, such as general-purpose or high-performance computing (HPC).

Behind the scenes Azure Virtual Machines runs on Hyper-V (see 2.2) and provides a set of features such as live migration, snapshot management, and auto-scaling [51].

Google Compute Engine

Google Compute Engine is a service provided by Google Cloud to manage computing resources in the cloud [52]. Compute Engine consists of three major parts: VMs, storage, and networking, where VMs are the core component.

VMs are offered in a wide range of configurations, such as standard, high-memory, and high-CPU, making them suitable for different computational workloads [53].

Compute Engine uses KVM as its hypervisor (see 2.2), and provides many features such as live migration, snapshot management, and auto-scaling [52, 48].

Comparison

OpenStack, OpenNebula, and CloudStack are all examples of software that can be used to build both private and public cloud platforms. A private cloud is an on-premise cloud that is hosted within an organization and is, more often than not, dedicated to internal use. A private cloud, on the other hand, is a cloud platform meant to facilitate external users. A common denominator for these systems is that they are open-source and that they are highly customizable and extendable. As such, much of the configuration and management of the system is done by the user, and the system can be integrated with other systems to fit specific needs. This means that many of the conveniences such as preconfigured VMs and features such as live migration often require specific configuration and setup [54, 55].

Amazon EC2, Azure Virtual Machines and Google Compute Engine are examples of VM management systems in public cloud platforms. They are designed to be more user-friendly and to provide a more complete abstraction layer over the underlying hardware and hypervisors. In comparison, these technologies can be seen as complete setups of the open-source platforms previously mentioned, with all the configuration and management done by a provider. As such, many of the conveniences and features are provided out of the box and are perceived as hypervisor- and hardware-agnostic. This means that the user does not need to worry about the underlying hardware and hypervisors, and can instead focus on managing the VMs. This creates a trade-off where the user has less control and customization but instead gains a more user-friendly and complete system [54].

Both private and public cloud solutions for VM management are designed to be scalable and flexible and support a wide range of use cases (although private cloud solutions scale worse in practice. Which solution to use depends on the specific needs of the organization, and the level of control, customization needs, and the required security level of the system [55]).

Briefly introduce traditional VM management systems like OpenStack, CloudStack, and OpenNebula, focusing on how they implement these core

features.

2.4 Containers and Kubernetes: A Paradigm Shift

Besides the common cloud service models (IaaS, PaaS, and SaaS), another service model has emerged, called *Container as a Service (CaaS)* [56]. CaaS is a container-based cloud service model that has gained significant traction due to the lightweight and portable nature of a container, allowing for simple application bundling that can run in a vast range of environments [57].

Containers are often used in conjunction with a container orchestration platform, such as Kubernetes, to manage the life cycle of the containers including deployment, scaling, and monitoring. As such, they are often used to build, test, and deploy applications consistently. A common use case for containers is to run microservices, which is a software architecture that employs a set of small, independent and loosely coupled applications [57].

2.4.1 Containers Explained

Containers are a form of virtualization that offers a lightweight alternative to VMs. Unlike VMs, containers do not require a hypervisor to run and instead run on top of an OS and share the OS kernel with other containers. By not requiring a hypervisor, containers can run with less overhead than VMs, making their life cycle relatively fast and efficient compared to VMs [6, 58]. The differences between container and hypervisor virtualization are visualized in figure 2.6.

A container is created from a container image, which is a lightweight executable software package that can run a specific application including dependencies and data files. By encapsulating an application and its dependencies in a single package, the container image becomes portable and independent allowing for the container to run consistently across different environments. A container image is created by describing its parent image, such as an OS image, and the steps to install and configure the application and its dependencies in a file, such as a Dockerfile. A parent image can itself have a parent image, creating a chain of dependencies. Furthermore, to enable caching and efficient storage, a container image can be divided into a series of steps referred to as layers. Each layer represents a change to the parent image, such as adding a file or installing a package. As such, layers act as checkpoints and allow for in-

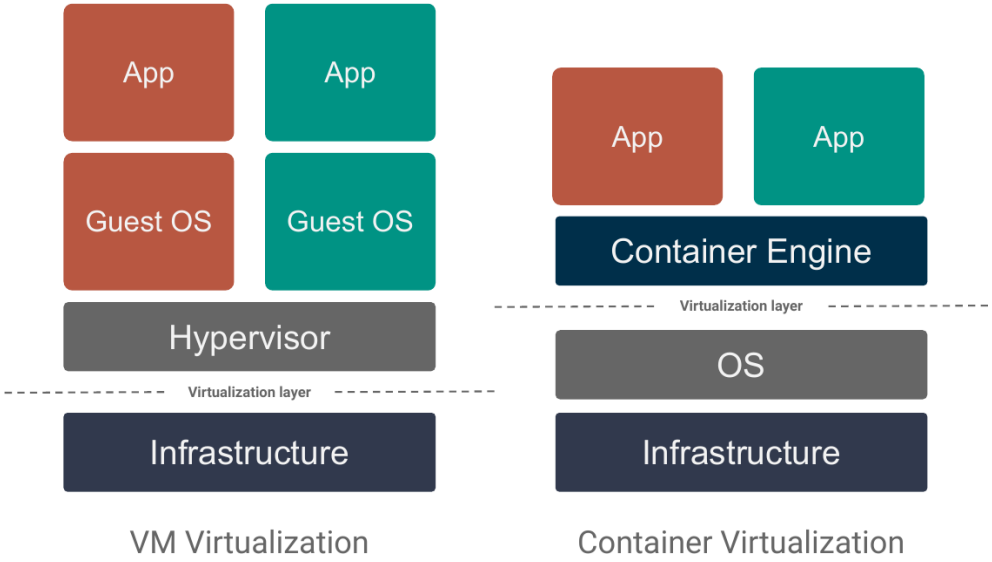

Figure 2.6: Overview of container and hypervisor virtualization

cremental builds where only the necessary layers are rebuilt when a change is made to the image [59, 60, 61]. Figure 2.7 visualizes the concept of container image layers, where each operation represents a change to the parent image.

Moreover, container images are often stored in container registries. A container registry provides central storage for container images and is used to distribute container images to environments that can run containers, such as a Kubernetes cluster. Common features for a container registry include versioning, access control, image scanning, and features to integrate with software development cycles, such as triggers to notify an external system when a new image is pushed to the registry [62, 63].

The Open Container Initiative (OCI) is a standardization effort that was created to ensure that containers are portable and interoperable across different platforms [64]. The OCI standard defines a runtime specification, image specification, and distribution specification, allowing for a container to be consistent in terms of its runtime environment, image format, and distribution method. Moreover, a container runtime, such as containerd, is used to create, deploy, and manage containers on a node [65]. Most runtimes, including containerd, implement the OCI runtime specification and can consistently run containers. This suggests that the OCI can be viewed as an abstraction layer, where the OCI runtime specification acts as an intermediary between the container runtime and the OS kernel, and the container image with the distribution specification acts as an intermediary between the container runtime and the container image

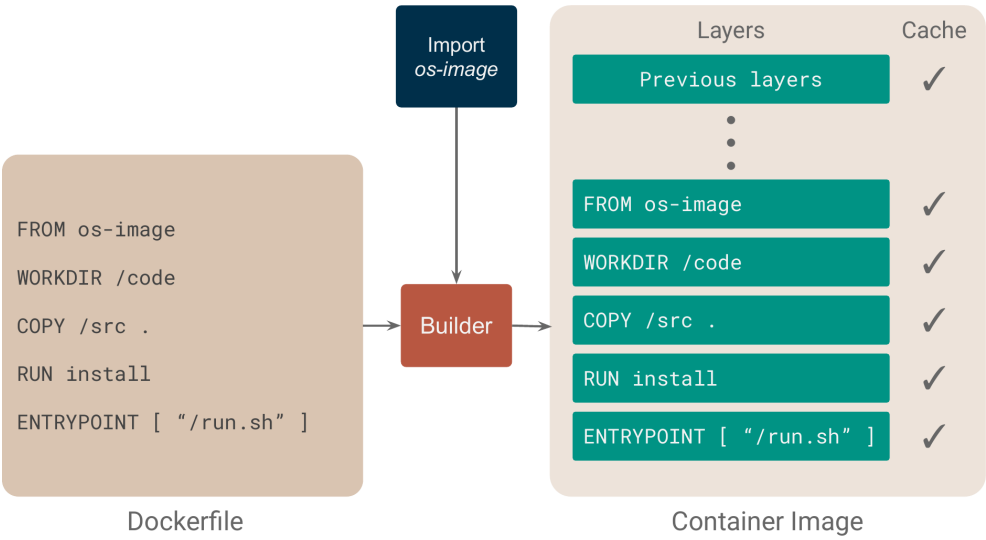

Figure 2.7: Overview of container image layers

registry.

2.4.2 Orchestration Using Kubernetes

Kubernetes is an open-source container orchestration platform that manages containers. It maintains a set of abstraction layers to manage the life cycle of containers and to provide a consistent environment for the containers to run in, facilitating tools for scheduling, scaling and networking [6].

Some of the abstractions used in Kubernetes to provide execution environments and communication include *Pods*, *Services* and *Deployments*. A Pod is the smallest deployable unit in Kubernetes, which groups one or more containers into a single unit that can be run on nodes. This means that Kubernetes ensures that all containers in a Pod are run on the same node. Encapsulating several containers allows for the logical grouping of coupled containers that share the same resources, such as networking and storage. This is commonly used when a Pod contains a main application container and a secondary container that provides logging and monitoring, called a sidecar container [66]. A Service represents a single-point-of-entry communication layer for a set of Pods and is used to provide a consistent way to access the Pods, such as load balancing. Services are especially useful in scenarios where Pods are fungible, meaning that the requesting client does not care which Pod it communicates with, as it creates an abstraction layer that hides the details of the Pods. This decouples frontends from backends and allows for a more scalable and resilient

system. Furthermore, services are also useful for rolling updates, as they can be used to ensure that a certain part of the traffic is routed to the new version of the application [67]. Contrary to Pods and Services, a Deployment is a higher-level abstraction on top of Pods that is primarily used for replication control. A Deployment does not represent an executable resource but instead describes a desired state in which a set of Pods should be. This includes the number of replicas and update strategy. As such, a Deployment proves useful when a set of Pods should be managed as a single unit, such as when scaling up or down, or when deploying a new version of an application [68, 69].

Furthermore, to support stateful applications Kubernetes mainly uses *PersistentVolumes*, *PersistentVolumeClaims* and *StatefulSets*. A PersistentVolume defines how and where persistent data is stored. Since there are many different ways to store data, such as local storage and network storage, a PersistentVolume is used to abstract the underlying storage. As such, each PersistentVolume is backed by a storage class, which defines the parameters and configuration to interact with the storage provider. A PersistentVolumeClaim, on the other hand, connects the PersistentVolume to a Pod. It acts as the bridge between the persistent storage and the execution environment effectively decoupling the Pod from the underlying storage. Contrary to PersistentVolumes and PersistentVolumeClaims, a StatefulSet is not used to manage the storage and is instead comparable to a stateful Deployment. A StatefulSet manages the life cycle of a set of Pods such that they are run predictably with their own unique identity. A common use case for a StatefulSet is scenarios that require each Pod to maintain a specific role, such as a primary or secondary database. A StatefulSet ensures that Pods maintain their role and that they are started in a predictable order, such as starting the primary database before the secondary database [70, 69].

Figure 2.8 presents a brief overview of the Kubernetes abstractions and how they interact with one another.

In certain scenarios, the built-in abstractions are not enough, and custom behavior is needed. For such scenarios, Kubernetes provides a way to extend its abstractions through Custom Resource Definitions (CRDs). A CRD is a form of extension for Kubernetes resources that allows for the creation of custom resources. It is used to extend the existing abstractions to provide additional features and capabilities that allow for a more feature-complete platform when custom behavior is needed. For instance, a CRD can be used to enforce certain fields to be specified to create the resource by applying custom validation rules [71, 69, 72].

There are many other abstractions in Kubernetes, each with its own set of

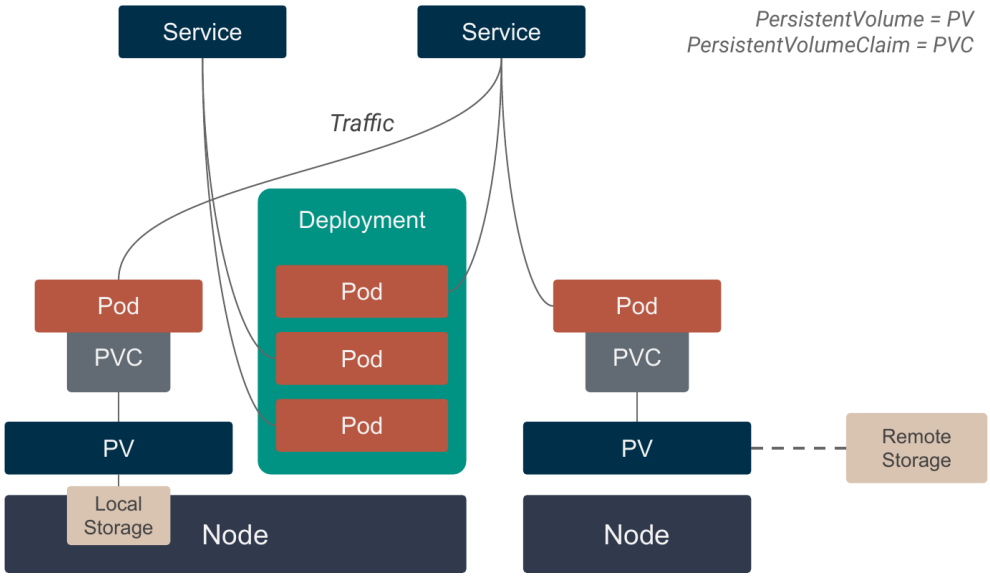

Figure 2.8: Overview of Kubernetes abstractions

features and capabilities that together create a feature-complete platform for container orchestration, but presenting them all in this report falls out of the scope of this project.

Kubernetes is inherently declarative, meaning that instead of issuing commands to perform a specific task, such as starting a container, Kubernetes is given a desired state for a set of resources, and it is responsible for ensuring that the current state of the resources matches a desired state. For this reason, Kubernetes provides auto-restart, re-schedule, and replication control to ensure a resource stays in the desired state. By specifying the desired state, an entire application can be contained in a single configuration file, including its container, networking, storage and security requirements [6, 69]. The entity responsible for ensuring the desired state is referred to as a controller, and there are many different types of controllers depending on the resources they manage. For instance, a Deployment controller is the entity responsible for ensuring that the number of Pod replicas in a Deployment matches the desired amount. Moreover, managing resources using a configuration file and a controller is referred to as the *Controller pattern*, and is a common design pattern in Kubernetes as it provides a consistent and predictable way to manage resources [73].

Architecture-wise, Kubernetes uses a client-server approach. The server, often referred to as the control plane, is responsible for managing the state of the cluster. It is responsible for scheduling, scaling and networking by manag-

ing system-related services such as the API server, scheduler, and controller manager. The API server can be viewed as the primary service for a Kubernetes cluster since it represents the gateway for all management tasks for the cluster by receiving and processing requests from the client. The client, often referred to as the worker node, is responsible for running the containers. The worker node consists of two primary components: the kubelet and the kube-proxy service. The kubelet is a service that wraps a container runtime and runs containers according to requests by the control plane [74, 69]. The kube-proxy service, on the other hand, manages the network communication for the containers. It can manage streams and forwarding rules for traffic using TCP, UDP or STCP [69, 75].

To interact with Kubernetes from the perspective of a user, a client is used to send requests to the API server. The client can be a CLI, such as kubectl, a web-based user interface, or a programmatic interface such as a REST API. The CLI tool kubectl is the most common way to interact with Kubernetes and is used to create, delete, and manage resources in the cluster [69].

Through its abstractions, life cycle management and declarative client-server architecture, Kubernetes provides a consistent and feature-rich platform for container orchestration. This has led to wide adoption of Kubernetes, and as such, it is often used in large-scale cloud environments [16].

2.5 Containers vs. Virtual Machines

Virtualization has enabled the commoditization of computing resources due to the benefits of being able to map virtual hardware to physical hardware (see 2.1) and has been key in enabling cloud computing. It has enabled the creation of flexible and scalable computing environments and has been used to build and manage large-scale cloud platforms. Virtualization has mainly brought two technologies: hypervisors and containers. Hypervisors are used to create VMs, including all associated aspects such as an OS. Containers, on the other hand, are used to create a lightweight and portable environment for an application to run in and are often run inside VMs. As such, a container comes with less overhead when compared to a hypervisor. However, by its nature, a VM can provide a more isolated environment than a container, leading to a more secure environment. Thus, the benefits of containers do not come without a cost and there is a trade-off between the two technologies; one is lightweight, portable, and quick to start while the other is more independent and secure [6, 76].

2.5.1 Comparative Overview

While containers and hypervisors are both technologies brought by virtualization, they operate on distinct levels of abstraction. Containers operate on a higher level than VMs using OS-level virtualization, which leads to the OS kernel with other containers. A hypervisor on the other hand can virtualize the entire hardware stack, including the entire OS, depending on the type of hypervisor used (see 2.2). As such, the two technologies have different use cases and are often used in conjunction with each other [76].

The main difference between containers and VMs is the level of isolation they provide. Containers share the OS kernel with other containers while VMs run on their OS. While this means a container can share resources more efficiently, it also means that a container is subject to a larger attack surface, as a vulnerability in the OS kernel can affect all containers running on the same host. VMs, on the other hand, reduce the attack surface by running more independently, meaning a compromised VM is less likely to affect other VMs running on the same host [77].

Another aspect of isolation is how co-located containers and VMs can share resources. When sharing resources such as CPU and memory, containers are vulnerable to the noisy neighbor problem. The noisy neighbor problem occurs when a resource-sharing mechanism, such as a scheduler, is unable to provide a fair share of resources. Some scenarios include when a container can consume all available resources due to a heavily loaded application or when a container intentionally consumes all available resources to starve other containers. Solutions exist to mitigate the noisy neighbor problem, including limiting the number of resources a container can consume through OS-level features such as *cgroups*. A cgroup is a feature in the Linux OS designed to limit, prioritize and isolate resource usage of a set of processes [78, 79, 80]. VMs, on the other hand, can provide a fair share of resources, as they operate in a more isolated environment, which leads to more resilient and predictable performance. However, the isolation does not mitigate the noisy neighbor problem for all resource types. For instance, a VM is still vulnerable to the noisy neighbor problem when sharing network-related resources such as network bandwidth and network-based storage [76].

Overcommitment is a technique used to improve packing efficiency, often referred to as *consolidation ratio*. Containers use the overcommitment mechanisms built into the OS itself, such as CPU overcommitment by multiplexing and memory overcommitment through virtual memory and memory swapping. Hypervisors, on the other hand, map virtual hardware devices to

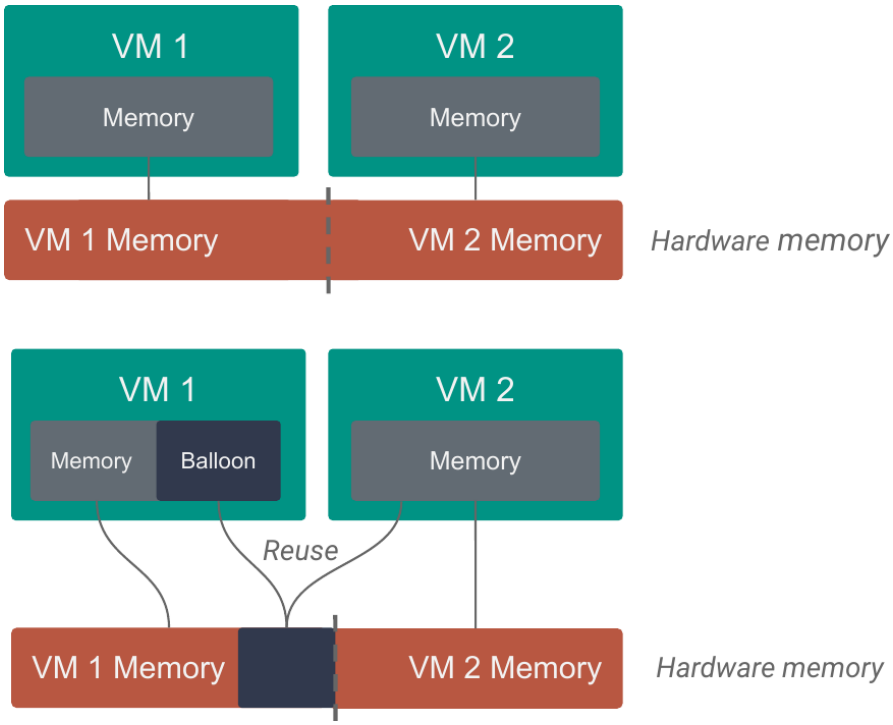

Figure 2.9: Overview of memory ballooning

physical hardware, creating virtual computing resources that are not directly tied to the physical resources. This works well for the CPU by creating virtual CPU (vCPU) cores. However, memory overcommitment is a more complex issue in hypervisors than in containers due to the way hypervisors allocate memory. Unlike containers, hypervisors assign a static amount of memory to a VM when it is created, and only reclaim the memory when the VM is deleted. Methods such as memory ballooning exist to address the limitation, which is a technique that treats memory as a shared resource across VMs and allows the hypervisor to reclaim memory from VMs when it is needed elsewhere. It is implemented as a driver that reserves unused memory (deflation) on a VM, rendering it unavailable to other processes running in the VM. The memory pages on the physical memory mapped to these pages in the VM are allocated to other VMs (inflation). This allows for VMs to share unused memory [81]. However, it is important to note that techniques like memory ballooning do not directly solve memory overcommitment for VMs, but instead reduce the need for it by managing the available memory more efficiently [76, 82]. Figure 2.9 visualizes the processes of deflating the memory of a VM and how it is used in another VM.

Live migration is a technique used to move a running resource from one host to another without interrupting the services provided by the resource. The technique is often used to balance the load or to perform maintenance on the hosting nodes. For VMs, it is a mature and widely technique for large data centers. Live migration works by copying memory pages from the source host to the destination host and then transferring the control of the VM from the source hypervisor to the destination hypervisor. Since VM disks are often stored on a shared storage, such as a Storage Attached Network (SAN), the VM disk is not transferred but rather reattached after the live migration is finished. For this reason, the VM can safely continue to run after its running state has been transferred to the destination host [83]. Live migration for containers, on the other hand, is not as mature as for VMs. Unlike VMs, containers require OS-level process migration, which is less reliable. However, several projects aim to provide live migration for containers, such as Checkpoint-restore in Userspace (CRUI). CRUI operates by checkpointing the state of a container, such as its memory and file descriptors, and then restoring the state on another host. While this is considered as container live migration, it is more comparable to snapshot and restore operations than a live migration operation, as CRUI freezes the container during the checkpointing, and converts the state of a container to a collection of files on disk, that can be used to restore the container on another host [84, 85, 86]. Another approach for live migrating containers involves terminating them on one host and restarting them on another. This strategy assumes the container is stateless or utilizes shared storage for its state, as is often used in Kubernetes since it does not provide container live migration. Instead, Kubernetes facilitates persistent storage through Persistent Volume Claims (PVCs), essentially allowing a user to map portions of the container's state to shared storage, and assumes the rest can be discarded when a container is terminated [69].

2.5.2 Decision Factors

The choice between containers and VMs depends on the specific use case, and there are several factors to consider when deciding which technology to use. Some of the core factors are security, scalability, and legacy support [76].

Security is a key factor when deciding between containers and VMs. Using containers means sharing the OS kernel, which implies that a vulnerability in the OS kernel could affect all containers running on the same host. Containers are thus, from an attack-surface perspective, more prone to some vulnerabilities than VMs, making VMs more suitable in scenarios that require a higher

level of isolation. For instance when an application manages sensitive or confidential data [87]. Moreover, sandboxing is a security mechanism that is used to restrict an execution environment such that it can safely run untrusted code without affecting the rest of the system. For instance, when analyzing the effect of malware a sandbox environment ensures that the malware is not able to escape the restricted environment. Sandboxing is done in multiple layers, including limiting the resources a process can access, such as memory and CPU, and limiting the system calls a process can make. As such, a sandbox environment requires a high level of isolation, which makes VMs a natural fit [88, 89].

Configuration drift is another aspect to consider regarding security. It is a phenomenon that arises when the configuration of a system changes over time due to several factors including manual intervention and software updates, and can lead to an undefined state in a system that could create security holes [90, 91]. Containers, as described in 2.5.1, are often run in a stateless manner using a single configuration file to describe the container, such as a Dockerfile. By being able to discard the state and recreate a container from a configuration file, the risk of configuration drift is reduced. As such, containers can provide a more consistent environment than VMs. Moreover, containers represent a more standardized and portable environment, using layers of dependencies, see Fig. By standardizing the environment, external tools are more easily able to provide security features such as vulnerability scanning [92].

When deciding between containers and VMs scalability is an important factor. One of the main benefits of containers is their ability to share the underlying OS. However, as described in 2.5.1, this also means that containers are more prone to the noisy neighbor problem, where co-located applications can cause performance interference. While this issue is related to isolation and security, it is more prominent in larger-scale environments, especially in multi-tenant environments [80, 92].

Scalability is also related to cloud-native applications, which are applications that are designed to run in the cloud [7]. A cloud-native application is characterized by being horizontally scalable, stateless, and resilient, and is often built using microservices, see 2.4. Cloud-native applications thus require small and lightweight environments, making containers with an orchestration platform such as Kubernetes a natural fit.

Legacy support is another factor to consider when deciding between containers and VMs. Both VMs and containers can provide environments for legacy applications, such as applications that require a specific OS version or a specific set of dependencies. However, certain legacy applications are not

able to run in a container due to the way they are designed, such as applications that require a specific OS kernel or a specific set of hardware. In such cases, a VM is often a better fit, as it enables virtualized hardware and a complete OS to be run [92, 93].

2.5.3 Towards Stronger Container Isolation

Containers are often used in conjunction with VMs due to the lack of strong isolation properties out of the box, see 2.5.1. However, several projects aim to provide the isolation properties of a VM and the lightweight properties of a container, such as Firecracker, Kata Containers, and gVisor.

Firecracker is a KVM-based hypervisor that creates lightweight VMs, called microVMs. Compared to a traditional VM, a microVM can start in a fraction of a second using only a fraction of the memory, comparable to that of containers. This is achieved through heavy optimization both in the hypervisor and in the guest functionality, such as reduced overhead through lightweight I/O and a stripped-down guest OS [8, 94].

Kata Containers is a project that, similar to Firecracker, aims to provide lightweight VMs that perform like containers while also providing strong isolation properties. Unlike Firecracker, Kata Containers uses both VMs and containers and replaces the container runtime, such as containerd (see 2.4.1) with the Kata runtime. The Kata runtime implements the OCI runtime specification to be interoperable with container ecosystems. However, instead of running the container directly on the host, the Kata runtime creates a VM, in which the container runs. This provides the isolation properties of a VM and the lightweight properties of a container. Kata Containers can achieve this by heavily optimizing the VM and its creation process towards running containers, as this is the only purpose of the VM [8].

Another approach to strong container isolation is gVisor, a user-space kernel that provides a secure environment for the container. Unlike both Firecracker and Kata Containers, gVisor does not use VMs to enhance the isolation properties but instead operates as an extension around the container. gVisor consists of two core components, *Sentry* and *Gofer*. Sentry is a user-space kernel that intercepts and handles system calls made by the container and can provide a secure environment for the container. Gofer is a file access component that provides the container with file access. Separating the file access from the syscall access allows for a more secure environment. For instance, a compromised Sentry does not imply read or write access to any file [95].

In summary, Firecracker stands out among the three since it does not use

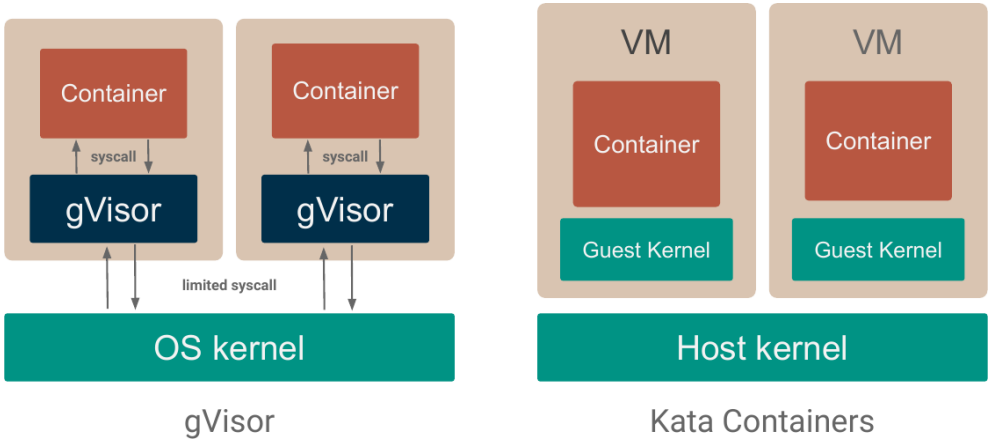

Figure 2.10: Overview of gVisor and Kata Containers

container runtimes. As such, Kata Containers and gVisor are more comparable. Figure 2.10 visualizes the differences between how Kata Containers and gVisor wrap a container to provide a more secure execution environment. However, due to the varying added overhead, they are yet to be widely adopted [8].

2.6 Extending Kubernetes with KubeVirt

KubeVirt is a novel project that aims to provide a unified platform for managing both containers and VMs using Kubernetes. It is designed to naturally blend in with the Kubernetes ecosystem and provide a consistent interface to manage the life cycle of VMs, such as deployment, scaling, and monitoring. KubeVirt being built on top of Kubernetes means that it does not implement all the components found in a traditional VM management system, such as the scheduler, but instead uses the mature and feature-rich services the Kubernetes ecosystem provides [96]. By using CRDs (see 2.4), KubeVirt can extend the Kubernetes API to provide a specification for VMs to support features commonly found in traditional VM management systems, such as live migration, scaling and snapshot management. Moreover, CRDs allow KubeVirt to enforce validation on the resources to prevent invalid configurations, such as a VM that references a non-existing storage disk [97, 98].

The design pattern chosen for KubeVirt's VM management is heavily inspired by other Kubernetes resources. A VM in KubeVirt is split into two resources called *VirtualMachine* and *VirtualMachineInstance*. The VirtualMachine resource is comparable to a StatefulSet and is used to describe the desired

state of a VM, such as the VM's desired running state, CPU and RAM requirements, as well as cloud-init configuration. The VirtualMachineInstance, on the other hand, is comparable to a Pod and is created by the VirtualMachine resource to represent the running state of the VM. As such, a non-existing VirtualMachineInstance represents a non-running VM, and an existing VirtualMachineInstance represents a running VM. The VirtualMachineInstance contains runtime information about the VM such as the VM's IP address, running state and resource usage.

Similar to other Kubernetes resources, such as Deployments and StatefulSets, KubeVirt uses controllers to manage the life cycle of the VMs. These controllers are together responsible for creating the environment in which a VM can be run, but also that the desired state of VirtualMachine resource matches the current state of its VirtualMachineInstance resource. For instance, when VirtualMachine is updated with requirements for CPU cores, the KubeVirt controllers are responsible for updating the VirtualMachineInstance to match the new specification [99, 97].

Live migration and snapshots are implemented as CRDs, allowing for a declarative interface to manage these features. This is in stark contrast to traditional VM management systems, where live migration and snapshot management are commonly implemented as imperative operations. For instance, by creating a LiveMigration resource, KubeVirt initiates a live migration operation for the referenced VirtualMachine resource [100, 101]. This is comparable to the binding operation process for a PersistentVolume and its corresponding PersistentVolumeClaim, where the binding takes place when a PVC is created [102]

An additional benefit of KubeVirt being built on top of Kubernetes is that KubeVirt-related resources can be managed using the kubectl CLI tool. This allows for the declarative creation, modification and deletion of VMs using the same interface as for other Kubernetes resources. However, certain operations are not possible to perform using kubectl. For instance, accessing serial or graphical consoles of a VM is not possible using kubectl. To enable these operations, KubeVirt provides a complementary CLI tool called *virtctl*. While virtctl primarily enables console access, it also provides convenience features such as live migration and snapshot management [103].

2.6.1 KubeVirt Technical Overview

On a technical level, KubeVirt's VMs are hosted inside Kubernetes Pods. As described in 2.4.2, a Pod is the smallest deployable unit in Kubernetes. Kube-

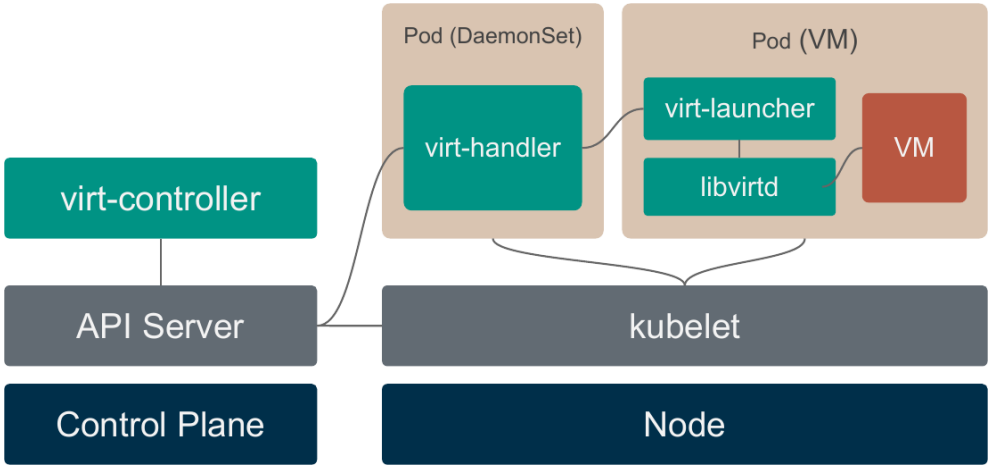

Figure 2.11: Overview of KubeVirt components

Virt does not create a new abstraction for an execution environment and instead encapsulates the VM within a Pod. This brings many benefits as it allows for seamless communication between the VM and the Kubernetes ecosystem since it complies with Container Network Interface (CNI). For instance, a VM can be exposed to the network using a Service or use PersistentVolumeClaims to provide persistent storage.

KubeVirt consists of four main components to enable the running of VMs in Kubernetes: *virt-controller*, *virt-handler*, *virt-launcher* and the *libvirt daemon*. The virt-controller and virt-handler are responsible for managing the life cycle of the VMs, where the virt-controller is responsible for creating the Pod in which the VM will be hosted, and the the virt-handler is responsible for creating the VM itself. The virt-launcher, on the other hand, manages the runtime environment for a VM, including starting, stopping and monitoring the VM. Finally, the libvirt daemon (libvirtd) provides an interface to manage the VMs hosted by the underlying KVM hypervisor. The relationship between the three components is visualized in figure 2.11, where the virt-controller operates closest to Kubernetes API-server, virt-handler operates as a DaemonSet on every node to communicate with the underlying hypervisor, and virt-launcher operates as a sidecar container in every Pod to manage the runtime environment of the VM [97, 99].

2.6.2 KubeVirt and GitOps

GitOps is a set of practices that enables using Git as a single source of truth for the desired state of the system, and it thus is heavily tied to the declara-

tive nature of Kubernetes, as the desired state of the system is described in configuration files. By managing the infrastructure in Git, resources can be version controlled, and any changes to the system can be tracked and audited. This creates a developer-centric workflow, where developers can manage the infrastructure using the same tools and processes they use for managing application code [104]. Moreover, since GitOps is used for systems to manage their resources declaratively, it is a natural fit for KubeVirt. As such, KubeVirt opens up the possibility of managing VMs using GitOps, meaning that VM specification and life cycle management can be version-controlled and audited [105]. Being easily integrated with GitOps further indicates that KubeVirt can unify the management of both containers and VMs using Kubernetes.

2.6.3 Comparison with Traditional Approaches

KubeVirt does not represent a new virtualization or hypervisor technology but instead leverages QEMU and KVM to run the VMs, which means KubeVirt is fundamentally similar to traditional VM management systems, such as OpenStack, CloudStack, and OpenNebula [106]. Therefore, KubeVirt can provide the same level of isolation and security as traditional VM management systems, as it uses the same underlying hypervisor technology. However, KubeVirt is not as feature-rich, as it is not designed to be a complete replacement for traditional VM management systems. For instance, OpenStack not only manages VMs and their life cycle but also provides additional services, including host provisioning and data warehousing [107]. KubeVirt, on the other hand, is primarily designed to manage VMs and their life cycle such that it can provide a unified platform for managing both containers and VMs. As such, KubeVirt integrates well with Kubernetes to provide a seamless experience for VM management. For instance, VMs created with KubeVirt support CNI plugins and integrate with PersistentVolumeClaims to provide storage [97, 108, 109].

2.7 Related Work

2.7.1 Current challenges

The Cloud Native Computing Foundation (CNCF) presents in a report some of the challenges of managing VMs using a traditional IaaS platform and uses OpenStack as an example [19]. While OpenStack has been a popular choice for managing VMs, it has been criticized for being complex and difficult to manage, as well as having a steep learning curve. They argue that the com-

plexity of OpenStack is not always needed and that it is possible to manage VMs using a simpler and more efficient platform, such as Kubernetes with KubeVirt.

Additionally, the CNCF report highlights the importance of having a unified platform for managing both VMs and containers, as it could reduce the complexity of the underlying technology stack and management costs. These claims are further supported by a report by Kubacki [20]. He argues that managing VMs using traditional VM management systems alongside Kubernetes leads to unnecessary costs, such as maintaining personnel and skill sets for two technologies. Moreover, by allowing the coexistence of VMs and containers in a single platform, KubeVirt could potentially reduce any barrier of entry around managing VMs for organizations that are already using Kubernetes for container orchestration, or act as a catalyst for organizations considering adopting Kubernetes.

2.7.2 KubeVirt challenges

While KubeVirt is presented as a solution to the challenges of managing VMs using traditional VM management systems, it is important to consider the challenges of using KubeVirt itself. A study by Trakulmaiphol and Piromsopa presents some shortcomings when migrating from a traditional VM management system to KubeVirt [110]. One of the issues identified is the management of IP addresses, particularly when maintaining static IP addresses in a dynamic IP environment, such as Kubernetes. In the study, they developed a framework to solve some of the issues on the application level, such as DNS queries, and provided a set of guidelines related to KubeVirt's unique features, such as the Multus Container Network Interface.

A common practice when managing Kubernetes clusters is to deploy Kubernetes nodes on VMs created by a VM management system. This means that when using KubeVirt, VMs are created inside other VMs leading to nested virtualization. Ye et al. argue in a study that nested virtualization raises security concerns due to the introduction of a larger trusted computing base (TCB) [111]. By using a larger TCB, the size of the system's attack vector increases leading to an increased risk for vulnerabilities. For this reason, nested virtualization might not be suitable when strict security is required, such as applications managing sensitive or confidential data. This study is particularly relevant as it highlights a potential limitation if KubeVirt is simply seen as another Kubernetes resource, and not as a VM management system.

2.7.3 Usability of Kubernetes and KubeVirt

Using Kubernetes as a VM management system is a relatively new concept, and there is limited research on the usability of KubeVirt. However, several studies have evaluated the usability of Kubernetes, which is relevant as KubeVirt is an extension built on top of Kubernetes.

Kubernetes API is inherently declarative, meaning an approach that promotes the usage of documents that describe a desired state of a system. A study by Sonninen investigates many of the possible benefits of using a declarative approach in system architecture [112]. The study found that a declarative approach facilitates GitOps, a way to manage infrastructure using files in Git, which simplifies system and resource management and enables easier rollbacks. Using Git as a source of truth for the desired state of the system has the added benefits of version control, audit and collaboration. These benefits were exemplified when CERN migrated the internal CMS project from a VM cluster to Kubernetes, as described in the two studies they published. The first study conducted by Imran et al. describes how the Kubernetes migration enhanced the overall system management compared to using a VM cluster [113]. While many of their experienced benefits stem from adopting containers instead of VMs, the study still suggests Kubernetes provided many advantages, such as reduced operational costs. The other study conducted by Badran presents the practical benefits they experienced by using Kubernetes and is more focused on the adoption of a declarative approach for system management [114]. It describes how using Kubernetes instead of a VM cluster allowed them to craft reusable and composable configurations that enabled their entire infrastructure to be managed as code.

The benefits of moving into Kubernetes are further exemplified in a study by Gelley [115]. It presents a case study where an organization moved its business application from a Platform as a Service (PaaS) provided by Cloud Foundry into Kubernetes. The study not only describes how Kubernetes provided tremendous flexibility in terms of configuration and deployment compared to the previous PaaS but also how the organization was able to leverage the Kubernetes ecosystem to provide additional features such as auto-scaling and monitoring. Moreover, by moving into Kubernetes, possible future migration to other platforms was made simpler due to how widely adopted Kubernetes is.

User experience in Kubernetes was investigated in a study by Packard et al. that gathered several scenarios in which projects were built and managed using Kubernetes [116]. Many aspects were considered in the study, including ad-

ministration such as managing persistent storage and limiting user resources with quotas, but also developer experience such as building and deploying applications. The study found that while system administrators face a learning curve when using Kubernetes, such as understanding the Kubernetes API and the ecosystem, users without prior knowledge of Kubernetes were able to successfully deploy applications. This finding is important to consider when evaluating KubeVirt, as it suggests that KubeVirt is a viable option even for users without prior knowledge of Kubernetes. Another study that investigated the user experience of Kubernetes was conducted by Boronin [117]. The study presents a case study where an organization faced a migration from a traditionally monolithic system architecture to a more cloud-native design. The study is relevant as it not only describes the challenges or benefits of Kubernetes itself but also considers the organizational impact of migrating to Kubernetes. This involves transferring skill sets among personnel, and the impact on the organization's culture.

In summary, these studies suggest that the benefits of using Kubernetes as a platform are not directly tied to what kind of resources are managed, but rather to the platform itself. Thus, when evaluating KubeVirt, it is important to not only consider benefits of KubeVirt itself but also the benefits that follow by adopting Kubernetes.

2.7.4 Performance Evaluations

Regarding performance, several studies have evaluated popular systems that can manage VMs. For instance, open-source alternatives such as OpenStack and CloudStack, or proprietary systems such as Amazon EC2.

A study by Kang et al. compared the performance of OpenStack and Amazon EC2 using a wide range of configurations and analyzed diverse aspects based on different hypervisor, storage and networking configurations [118]. Apart from the performance comparisons, the study also investigated performance impact when using more advanced features, such as Infiniband networking instead of Ethernet. The study showed mixed results, where OpenStack outperforms EC2 in some scenarios such as when using Infiniband networking, but EC2 tends to outperform OpenStack in parallel processing. As such, the study concluded that component-wise, each system has strengths. This study brings with it relevant perspectives as it highlights many of the aspects that are important to consider when evaluating the performance of VM management systems.

Another study by Mullerikkal and Sastri compared OpenStack and Cloud-

Stack regarding implementation complexity, overall stability and performance [119]. The performance tests were designed to measure disk- and network I/O, and not CPU or memory usage. The study found that while OpenStack had a more complex implementation, it was more stable. Furthermore, by using a set of benchmarks it found that OpenStack had a better overall performance compared to CloudStack. However, a few years later, a complementary comparison between OpenStack and CloudStack was conducted by Paradowski et al. [120]. The comparison focused on performance and used several benchmarks such as deployment time, deletion time, and CPU utilization during deployment and deletion, and revealed significant and quantifiable differences between the two platforms. These tests were designed to represent user interactions with the system. However, the scale of the tests used in the comparison was relatively small and did not explore how the systems manage VMs under higher load. Analyzing the performance of a VM management system under higher load is important, as it can reveal how well a system scales when the ratio between VMs and nodes increases.

In contrast, a later study by Amaral benchmarked KubeVirt uses a setup with a large number of VMs relative to the size of the nodes, aiming to explore the limits of KubeVirt's control plane (scheduler etc.) [121]. The study found that, while KubeVirt was able to create all VMs, the control plane suffered a heavy load, indicating a possible large overhead when KubeVirt provisions a large number of VMs. This finding is important to consider when evaluating KubeVirt, as it indicates that KubeVirt may not yet be suitable for managing a large number of VMs.

In terms of network performance, a study conducted by Kolahdouzan investigated the networking traffic pattern in KubeVirt and OpenStack by performing networking performance tests to evaluate potential VM overhead [122]. The study found that the network overhead was significantly larger for scenarios in which VMs reside on separate nodes. However, while the study found overhead for both KubeVirt and OpenStack, it found that OpenStack had a considerably larger overhead. While this project will not focus on networking performance, it is important to consider the findings of this study when evaluating KubeVirt.

Hypervisors are a key component of VM management systems, and several studies have evaluated the performance of hypervisors. A study by Hwang et al. evaluated the performance of a range of hypervisors including KVM, Hyper-V and Xen [123]. The study found that applications and assigned resources had a significant impact on the overhead of the hypervisors. However, the study concluded that no single hypervisor was the silver bullet and that

the choice of hypervisor should be based on the specific use case. Reviewing the performance of hypervisors is important when evaluating KubeVirt, as KubeVirt uses KVM as its hypervisor.

Chapter 3

Methods

3.1 Choice of Method

3.1.1 Performance Analysis

As presented in the literature study, Paradowski et al. present a set of benchmarks used to evaluate CloudStack and OpenStack [120]. This comparison will be used as the foundational basis for the performance analysis, using many of its approaches when testing a VM management system. For this project, KubeVirt will primarily be compared to OpenNebula. This decision was made together with Net Insight, to limit the scope to a single VM management system, as well as to ensure that the tests are relevant to Net Insight's current infrastructure. OpenNebula was chosen as it is currently used by Net Insight and is a popular open-source VM management system. *Other alternatives, such as CloudStack and OpenStack, will be considered if time allows it*.

The testing process will be conducted on Azure using a group of VMs that will act as *hosting nodes*. This approach was chosen to ensure that the tests are not affected by the underlying hardware and to facilitate reproducibility.

A set of scripts will be created to automate the testing process, including creating nodes in Azure and then installing and configuring a VM management system on the nodes. After the tests are finished, the results will be collected by the scripts, and the nodes will be deleted in Azure. The scripts will thus be used to create a complete setup for each system in a consistent manner using a single command. The programming language Go will be used to create the scripts since it is commonly used to interact with APIs.

In the study by Paradowski et al., they used several preconfigured VM types in the tests, including *tiny*, *small*, and *medium*. As such, this project will use a

similar approach, with CPU ranging from 1 to 4 CPU cores, and RAM ranging from 128 MB to 4 GB (see Table 3.1.1). Furthermore, VMs will use the lightweight Alpine Linux image. This helps improve the accuracy of tests, as a lightweight OS image is less likely to introduce noise in measurements [124]. It also means VMs can be smaller, allowing for more VMs to be created per node and quicker test runs since a lightweight image is usually faster to boot than a larger image.

Each test case will use a setup of three nodes, one acting as the management node, and the other two as worker nodes. Having more than one worker node is crucial since live migration tests involve moving VMs between nodes. The nodes will be configured to be large enough to facilitate a broad range of tests, including tests with many small VMs and tests with a few large VMs. For this, Azure VM type A3 will be used. This VM type has 4 CPU cores and 7GB of RAM, enabling it to run many of the VMs that will be used in the tests. The price of the A3 VM type is 0.2 USD per hour, which was deemed acceptable for the project [125].

Below is a summary of the metrics that will be used to evaluate the performance of the VM management systems. The metrics will be used for the affected node in the test, for instance, when live migrating VMs it is important to measure both the source and the destination node, while when a VM is created, it is only necessary to measure the node where the VM is created. However, the control-plane node will be measured in all tests, as it is the node that is responsible for scheduling and managing the VMs.

- Metrics
  - CPU usage
  - Memory usage
  - Disk usage
  - Duration

Below is a summary of the tests that will be performed. They can be divided into three categories: basic, limit testing, and scaling. The basic tests are designed to evaluate the basic functionality of the VM management systems, such as creating and deleting VMs. The limit testing tests are designed to evaluate how a VM management operates when the maximum capacity is reached in the worker nodes. 100 tiny VMs were specifically chosen to reach maximum RAM capacity for two worker nodes. The scaling tests are designed

to evaluate how a VM management system handles the scale-out and scale-in of worker nodes, both with and without VMs on it (that would need to be migrated). Each test will be run 10 times.

- Tests
  - Create each type of VM and delete it
  - Create 100 tiny VMs and delete them
  - Create VM Snapshot and delete it
  - Live Migrate a VM
  - Scale from 2 to 10
  - Scale down 10 to 2 workers without VMs
  - Scale down 10 to 2 workers with 1 small VM on each worker

The following is a summary of the setup:

Role	CPU	RAM	Disk
Control Plane (Azure A3)	4	7GB	120GB
Worker 1 (Azure A3)	4	7GB	120GB
Worker 2 (Azure A3)	4	7GB	120GB
Scale-up Workers 3-10 (Azure A3)	4	7GB	120GB

Table 3.1: Node specifications

Type	CPU	RAM	Disk Size
Tiny	1	128MB	1GB
Small	1	1GB	1GB
Medium	2	2GB	10GB
Large	4	4GB	100GB

Table 3.2: VM specifications

3.1.2 Usability Analysis

To accurately measure the usability of KubeVirt, a set of tasks has been crafted. These tasks have been designed in cooperation with a system administrator at Net Insight. The tasks are designed to be representative of the tasks that a system administrator would perform when managing VMs, including creation, deleting, and debugging.

Below is a summary of the tasks that will be performed. The tasks will be performed by a group of system administrators at Net Insight using the tools *kubectl* and *virtctl*, and not a graphical user interface [126]. This approach was chosen as the system administrators at Net Insight are familiar with using the command line interface (CLI). Moreover, the VM specifications will be configured uniformly across all tasks using 1 CPU and 512 MB RAM and will use the lightweight Alpine Linux image. This is to ensure that the VMs are fast to create and start, and to emphasize the usability of KubeVirt and not the performance of the VMs.

- **Basic usage**
  This is a basic task meant to act as a starting point for the usability analysis. KubeVirt offers different ways to deploy VMs depending on the use case. For instance, using a *containerDisk* (stateless) versus using a *Persistent Volume Claim* (stateful), or applying cloud-init scripts using *cloudInitNoCloud* [127]. The system administrator will be presented with brief descriptions of the different methods, as well as links to the official KubeVirt documentation, and will be asked to create a VM using the method they find most suitable. After creating a VM, the system administrator will be asked to perform basic operations including starting, stopping, and restarting the VM, as well as accessing the VM using SSH and VNC, and then finally deleting the VM.

  After the task is finished, the system administrator will be asked to provide feedback on the experience by answering the following questions:

  - How easy was it to decide which method to use for creating a VM?
  - How intuitive was the process of creating and deleting a VM?
  - How would you describe the process of starting, stopping, and restarting a VM?
  - How straightforward was it to access the VM using SSH and VNC?

  - Describe, if any, the issues you encountered during the task. Were the issues related to KubeVirt, documentation, or other?

- **Maintenance**
  This task is designed to evaluate how well KubeVirt supports tasks related to maintenance. The first part of this task includes creating a snapshot of a preconfigured VM, after it is finished, the system administrator will asked to restore the VM from the snapshot. The second part of this task is migrating a VM from one node to another. The system administrator will be presented with a VM that is running on a node and will be asked to migrate the VM to another node. Links to KubeVirt's official documentation will be provided as a reference.

  In this task, the preconfigured VM will use a containerDisk, as it provides a stateless VM that is reinstalled at every boot [127]. By reinstalling the VM at every boot, the system administrator can easily reset the VM to a known state, which prevents any issues related to the VM's state.

  After the task is finished, the system administrator will be asked to provide feedback on the experience by answering the following questions:

  - How straightforward was it to create and restore a snapshot of a VM?
  - How intuitive was the process of migrating a VM from one node to another?
  - Describe, if any, the issues you encountered during the task. Were the issues related to KubeVirt, documentation, or other?

- **Debugging**
  This task is designed to evaluate how well KubeVirt supports debugging. KubeVirt uses API validation to ensure that VMs are correctly configured before creating them [128]. However, it is still possible that a VM is not starting if the misconfiguration is validated at runtime. For this reason, a system administrator will be presented with a VM that is not accessible due to cloud-init misconfiguration that prevents SSH access to the VM but is accessible using VNC. The system administrator will be presented with the basic problem statement "The VM is only accessible through VNC and not SSH" and will then be asked to debug the issue to fix the misconfiguration. Links to KubeVirt's official documentation will be provided as a reference.

In this task, the preconfigured VM will use a containerDisk, as it provides a stateless VM that is reinstalled at every boot [127]. By reinstalling the VM at every boot, the system administrator can easily reset the VM to a known state, which is useful when debugging.

After the task is finished, the system administrator will be asked to provide feedback on the experience by answering the following questions:

- How easy was it to debug and find the issue?
- How easy was it to fix the issue?
- Does the debug process differ from debugging a traditional VM management system?
- Describe, if any, the issues you encountered during the task. Were the issues related to KubeVirt, documentation, or other?

After the tasks are finished, the system administrators will be asked to provide overall feedback by answering the following questions:

- How would you compare the experience of using KubeVirt with other VM management systems you have used?
- How intuitive is the workflow of KubeVirt? Can you use it without reading the documentation?
- How would you describe the overall experience of using KubeVirt?

System Management

The last factor that will be considered in this project is system management. This includes exploring how the system is managed and is mainly concerned with integrations with other technologies to understand how well KubeVirt integrates with other systems. For instance, how well KubeVirt integrates with GitOps, a way to manage infrastructure using files in Git [112, 129]. Exploring system management also involves exploring how well VMs in KubeVirt can communicate with other Kubernetes resources, such as Pods, which reflects how well it can integrate with container-based applications. This will be evaluated through literature and documentation study [129]

Chapter 4

Results

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like "Huardest gefburn"? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.

Chapter 5

Discussion

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.

Chapter 6

Conclusions

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.

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Appendix A

Something Extra